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MIL-F-16377/53(SHIPS)
31 August 1971
SUPERSEDING
MS17231(SHIPS)
30 December 1968
(See page 24)

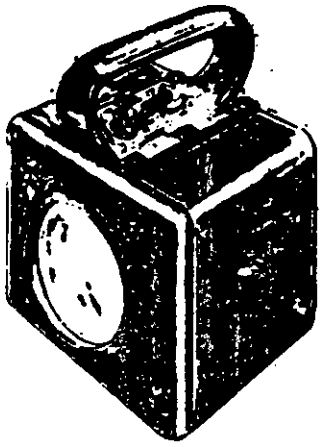
MILITARY SPECIFICATION SHEET

FIXTURES, INCANDESCENT, DETAIL LIGHTING,

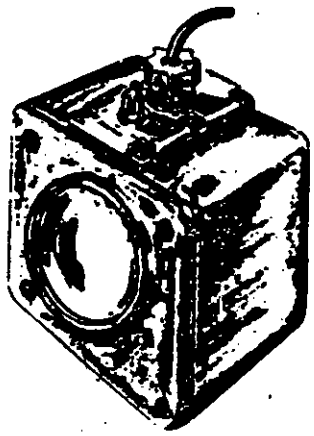
LANTERN, HAND, PORTABLE AND RELAY WATERTIGHT

SYMBOLS 100.2, 101.2, 102.2 and 108

The complete requirements for procuring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).



PORTABLE



RELAY OPERATED

Figure 1 - Lanterns, hand, portable and relay.

All dimensions in inches.
For tolerances other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6230

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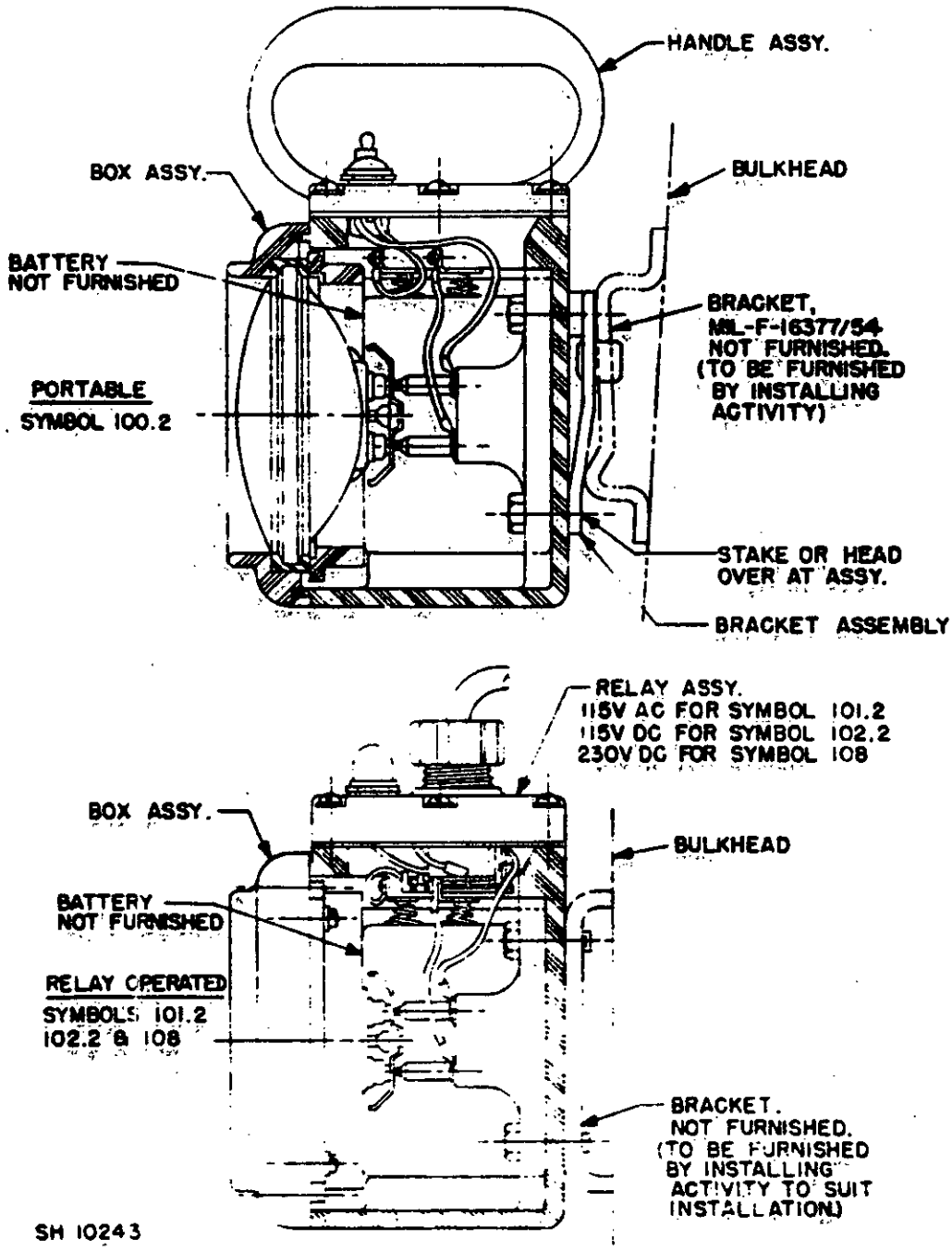


Figure 2 - Lantern assembly and typical mounting.

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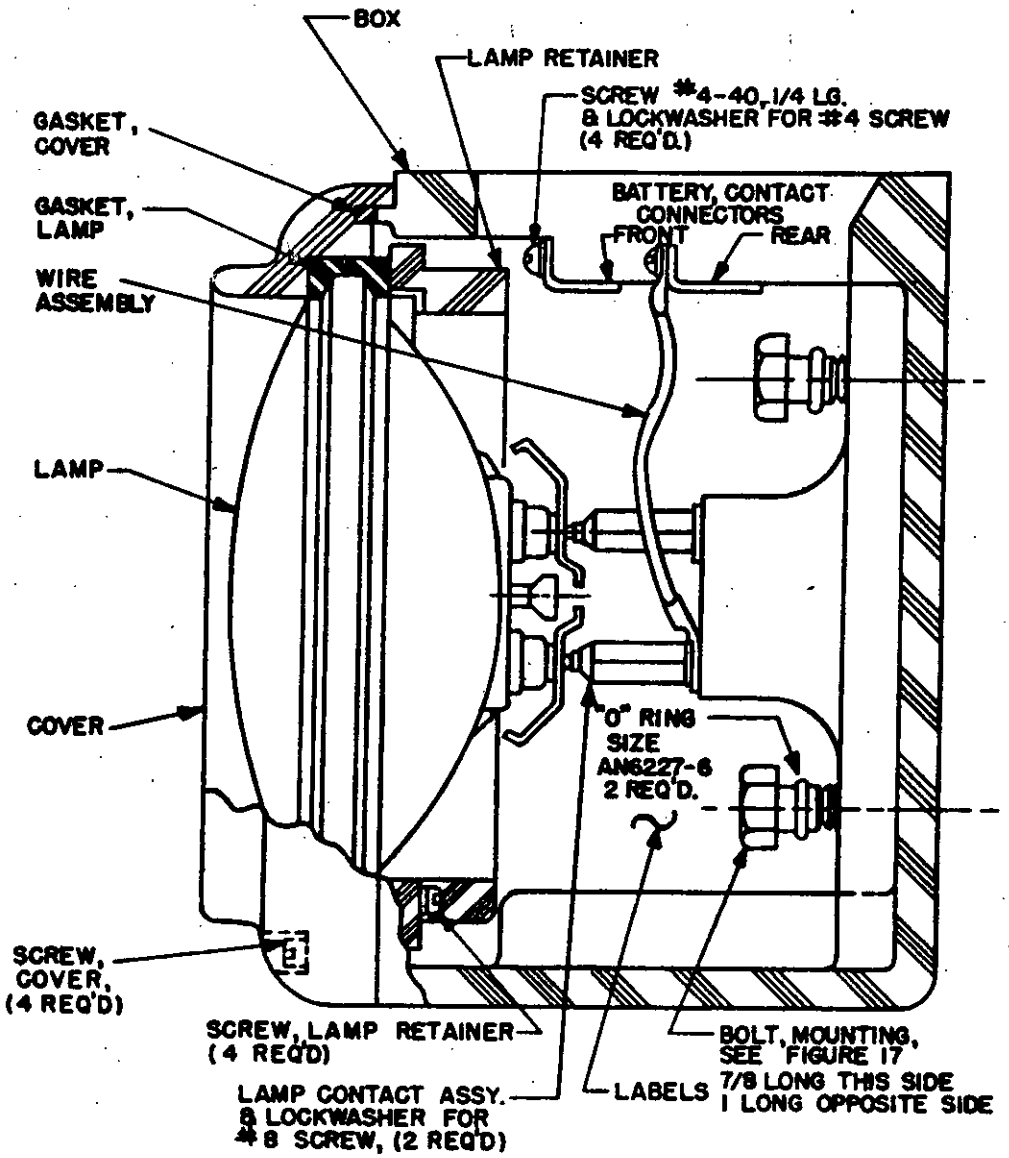
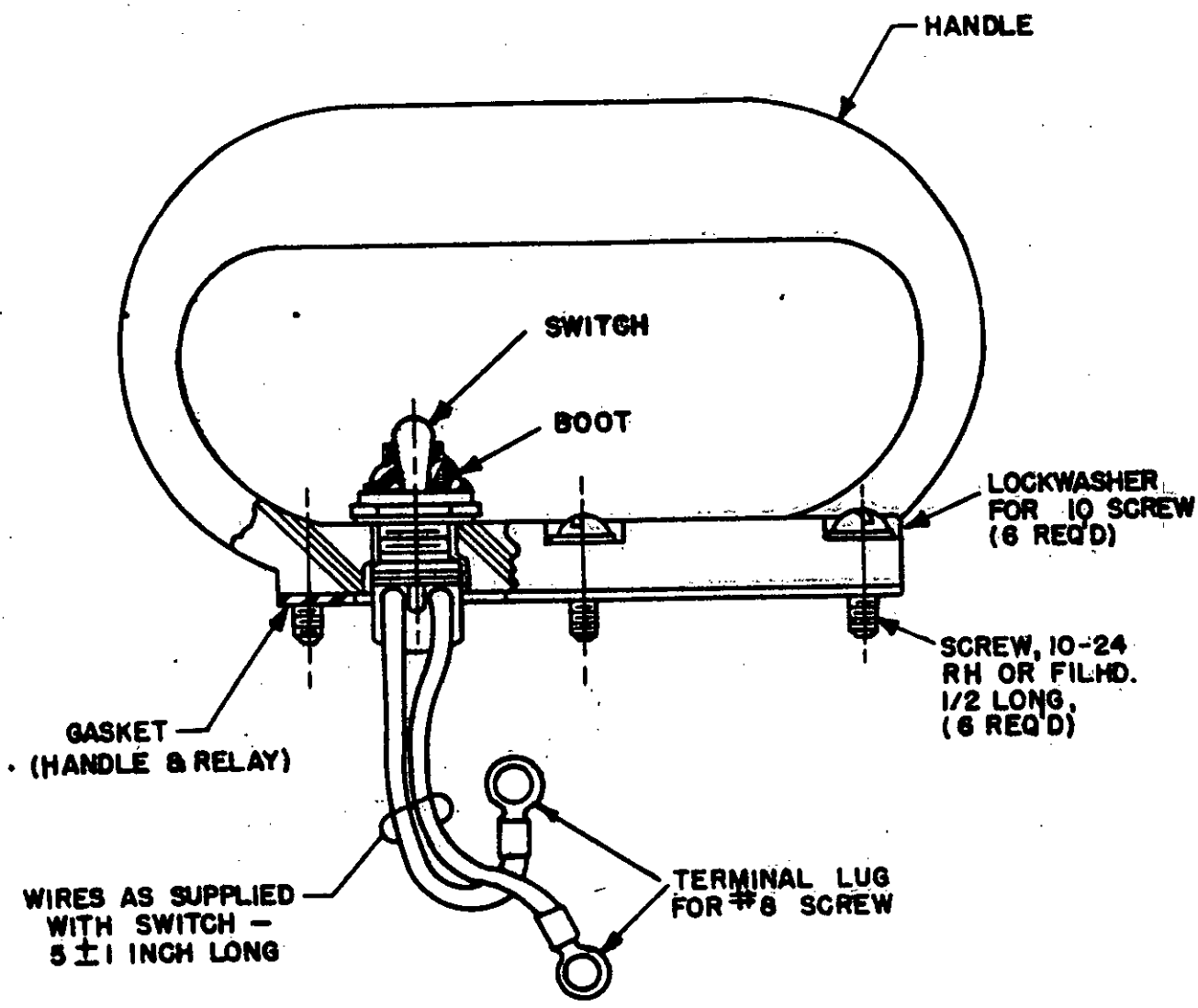


Figure 3 - Body assembly.

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SH 10245

Figure 4 - Handle assembly.

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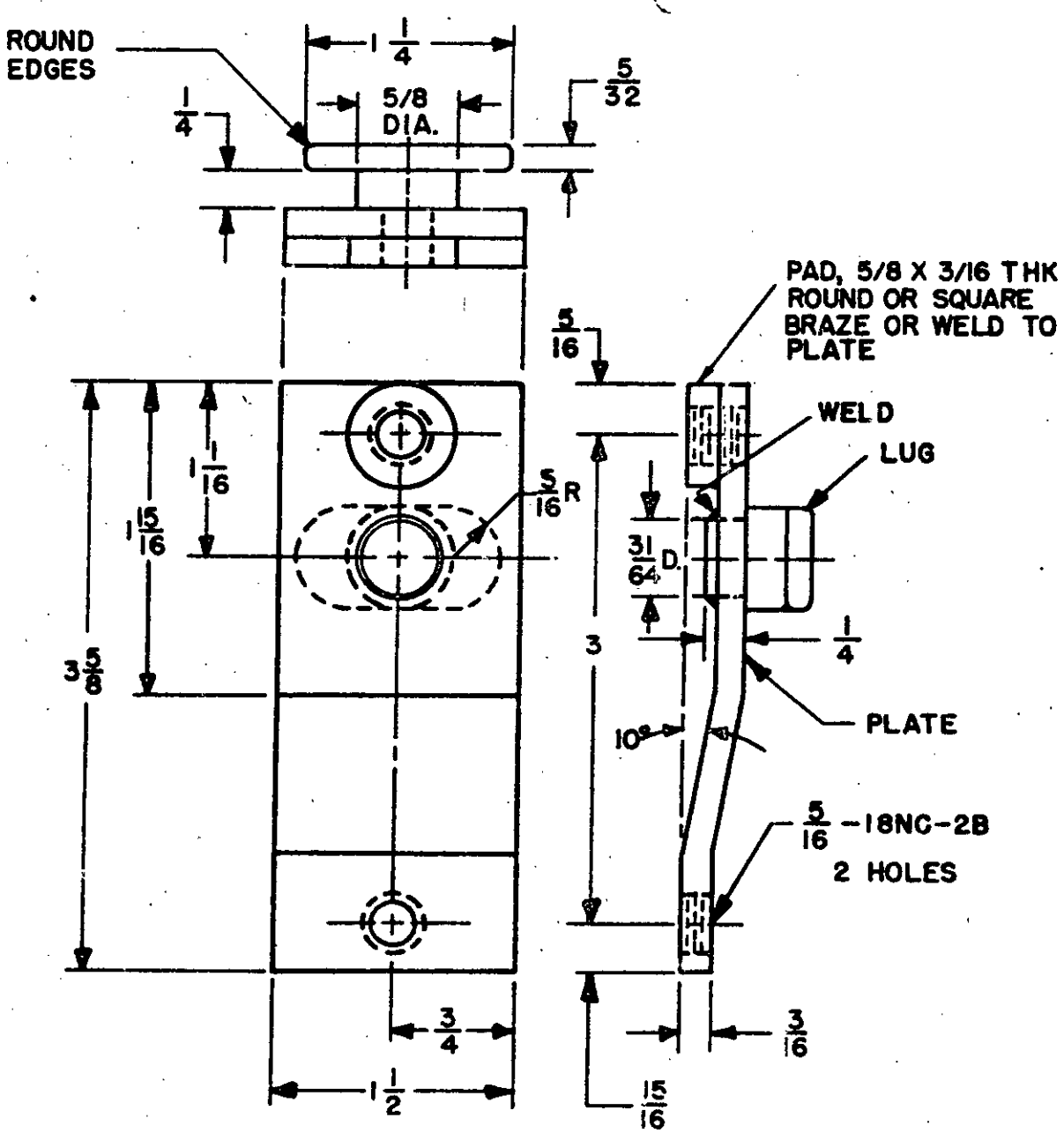


Figure 5 - Bracket assembly.

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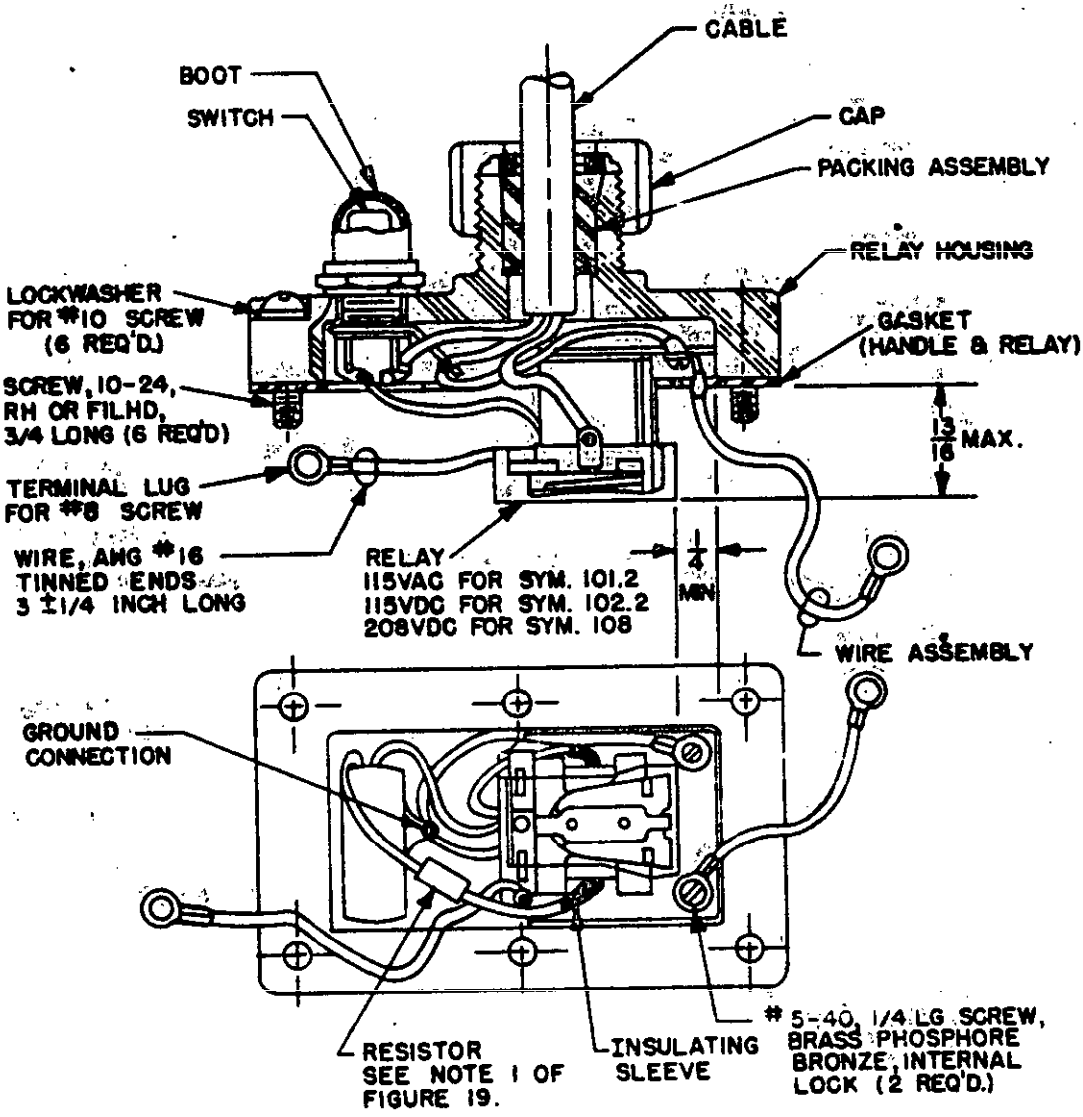
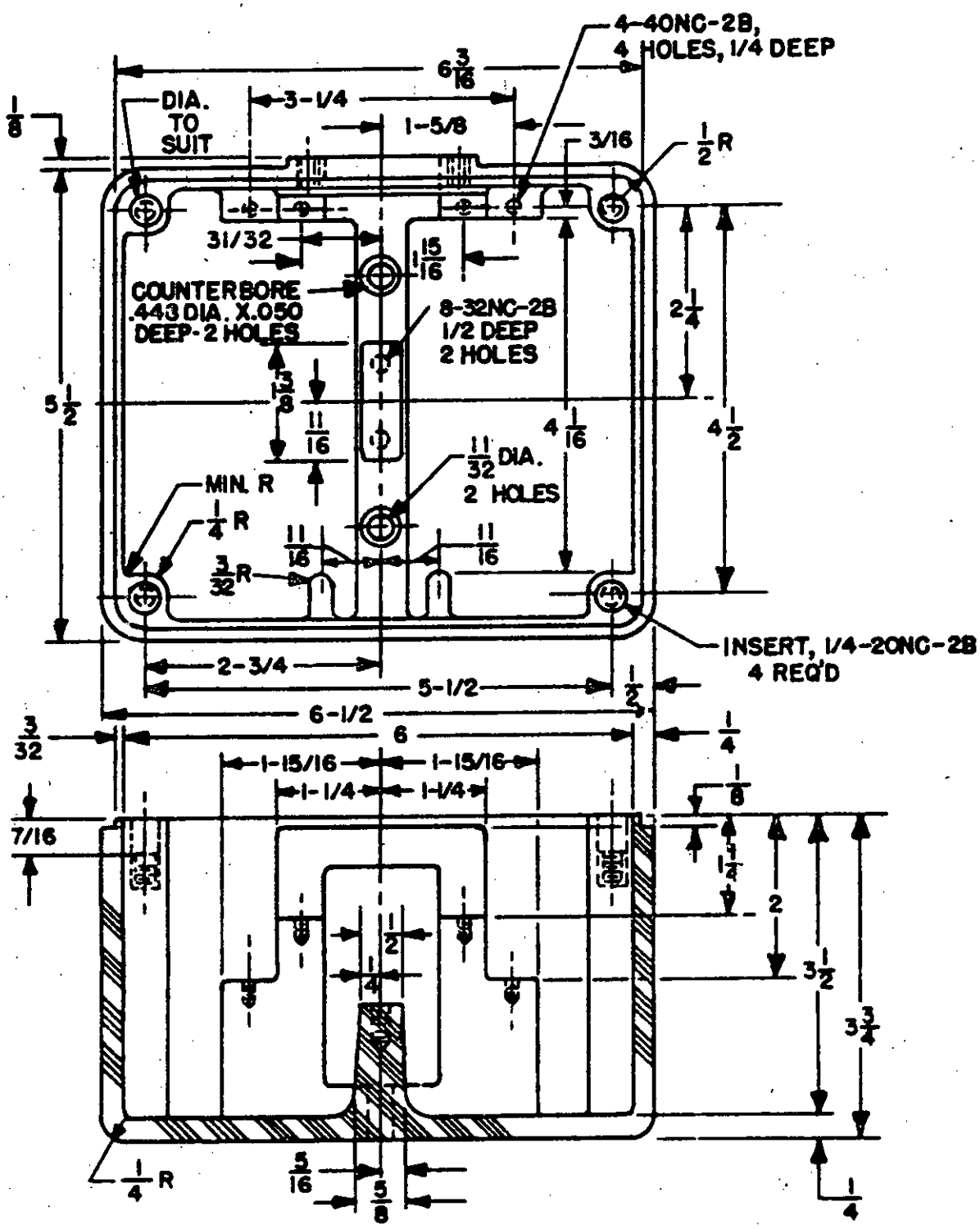


Figure 6 - Relay assembly.

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SH 10248

Figure 7 - Box.

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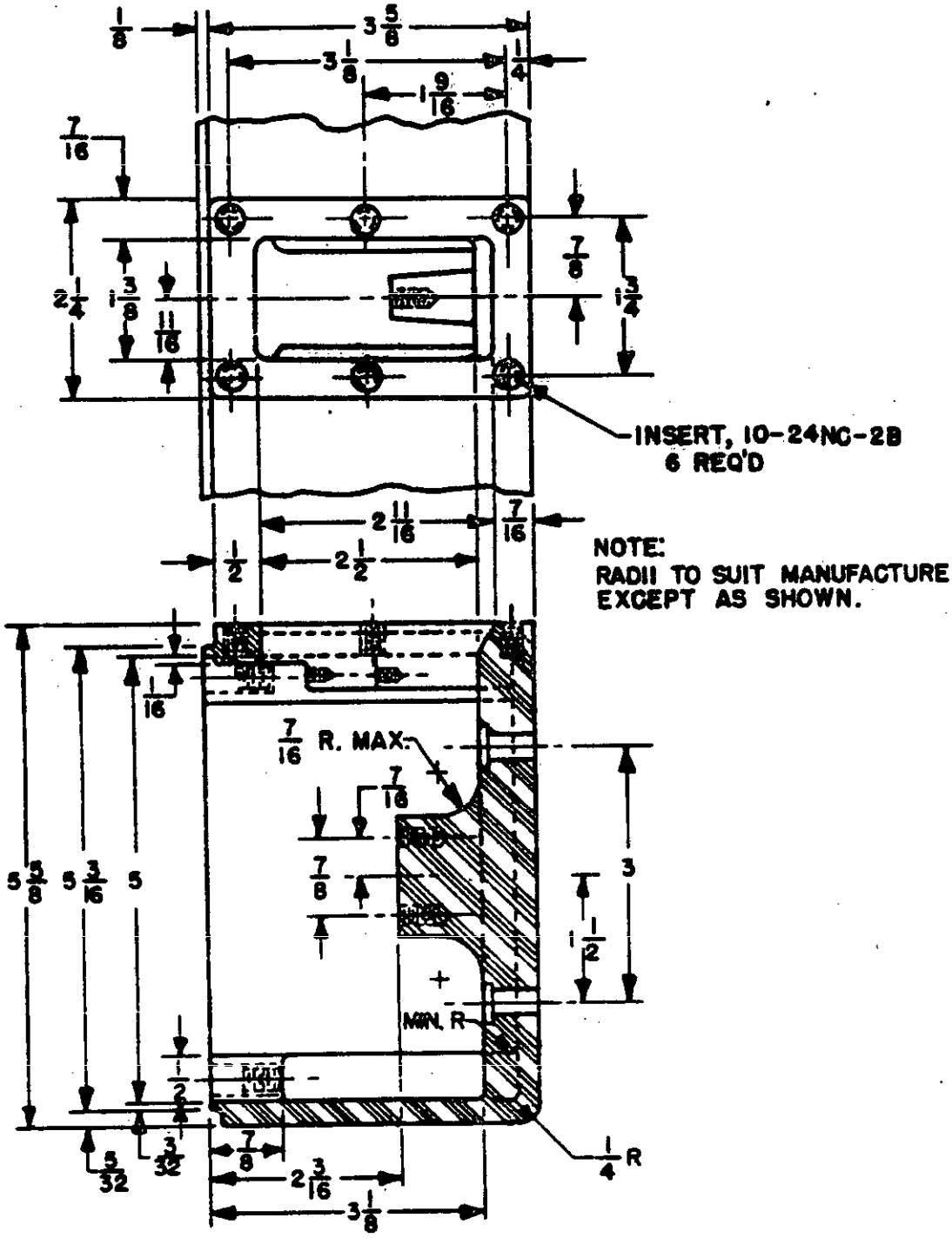


Figure 7 - Box (Cont'd).

SH 10248

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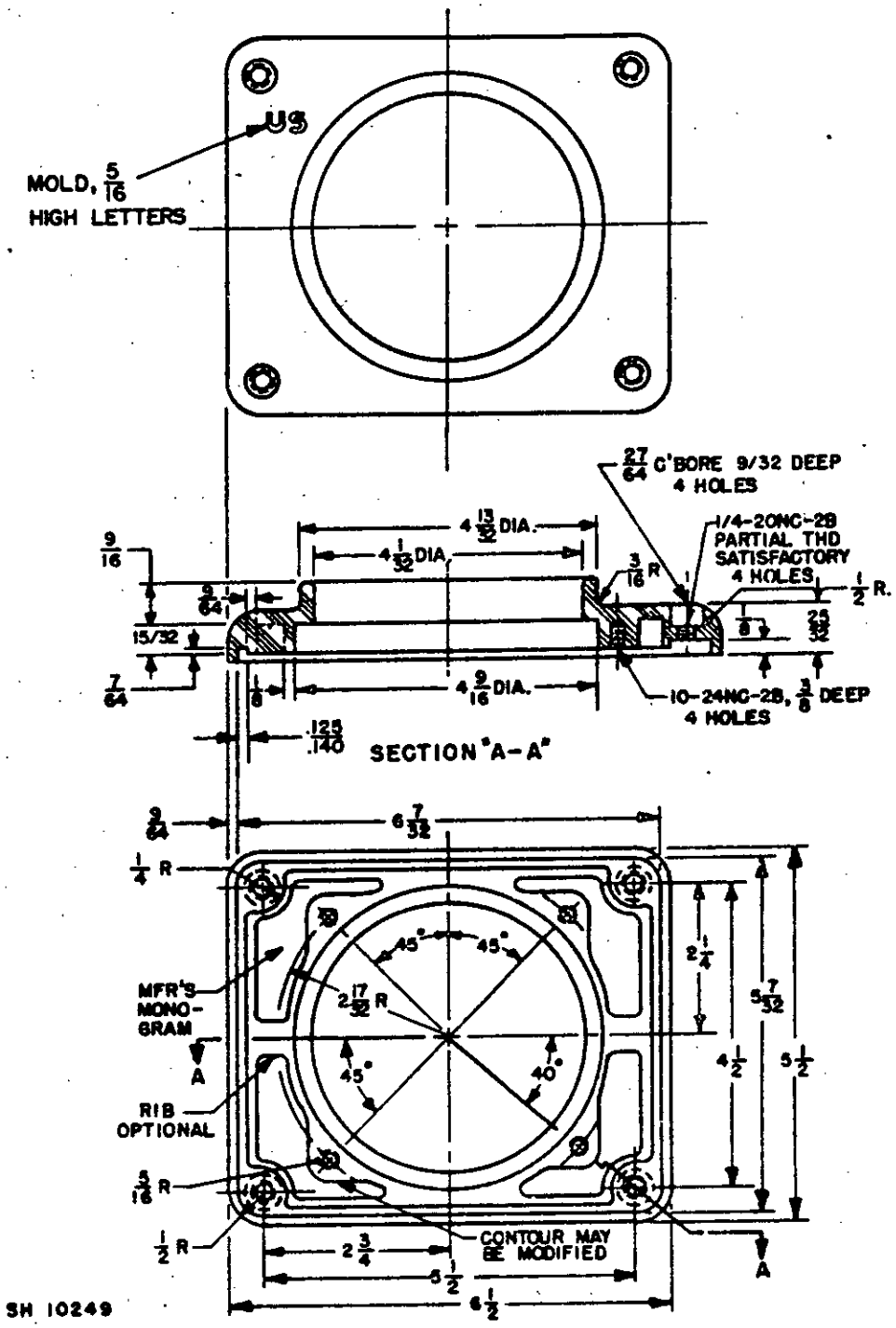
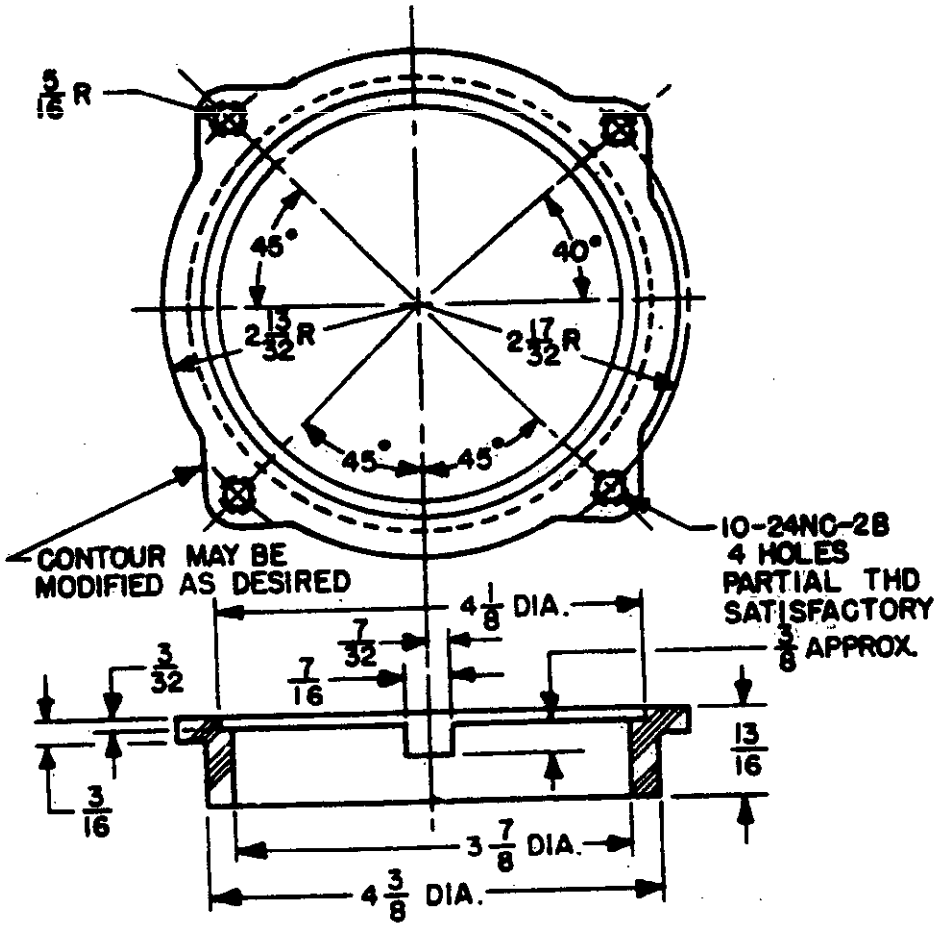


Figure 8 - Cover.

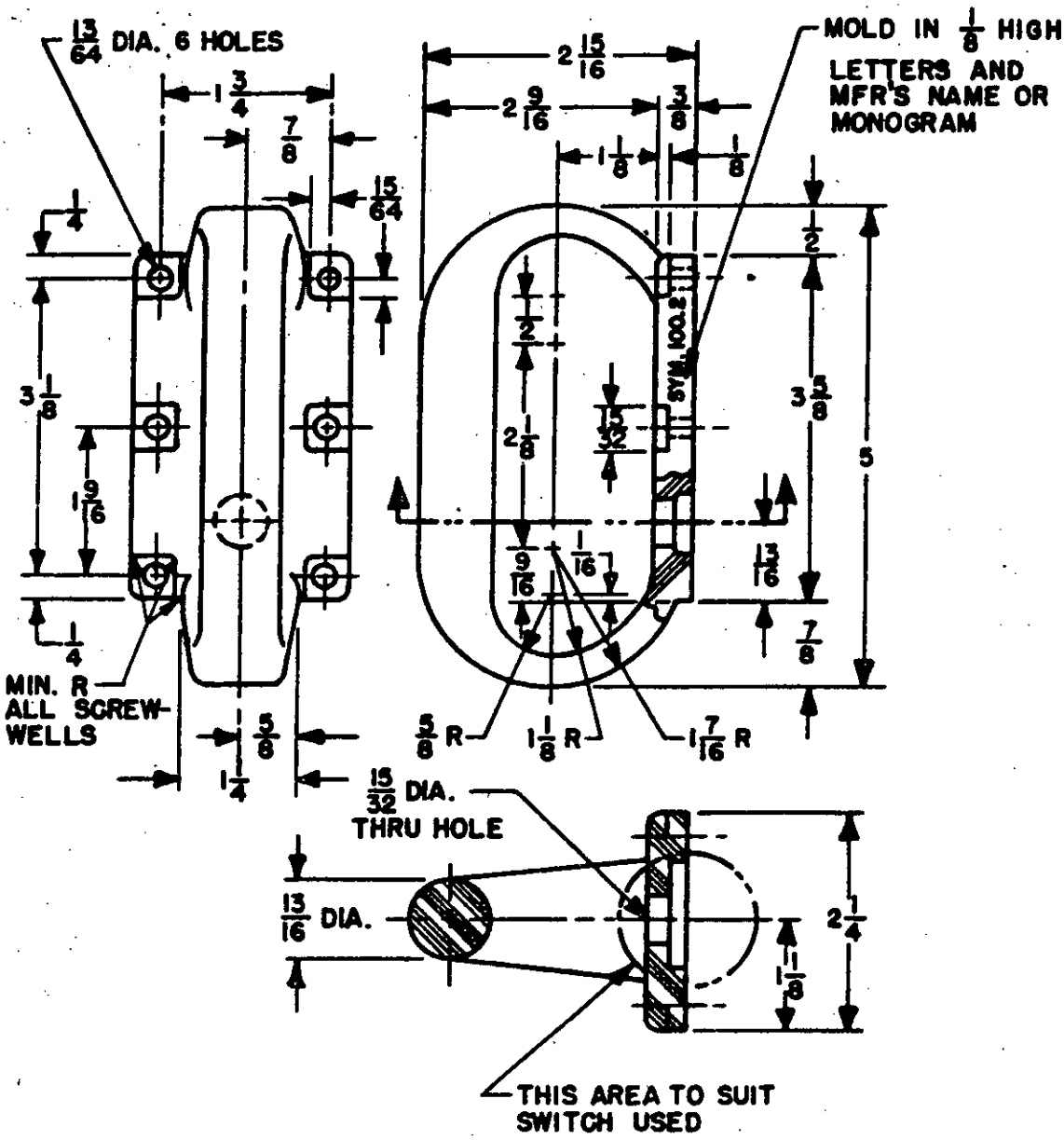
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SH 10250

Figure 9 - Lamp retainer.

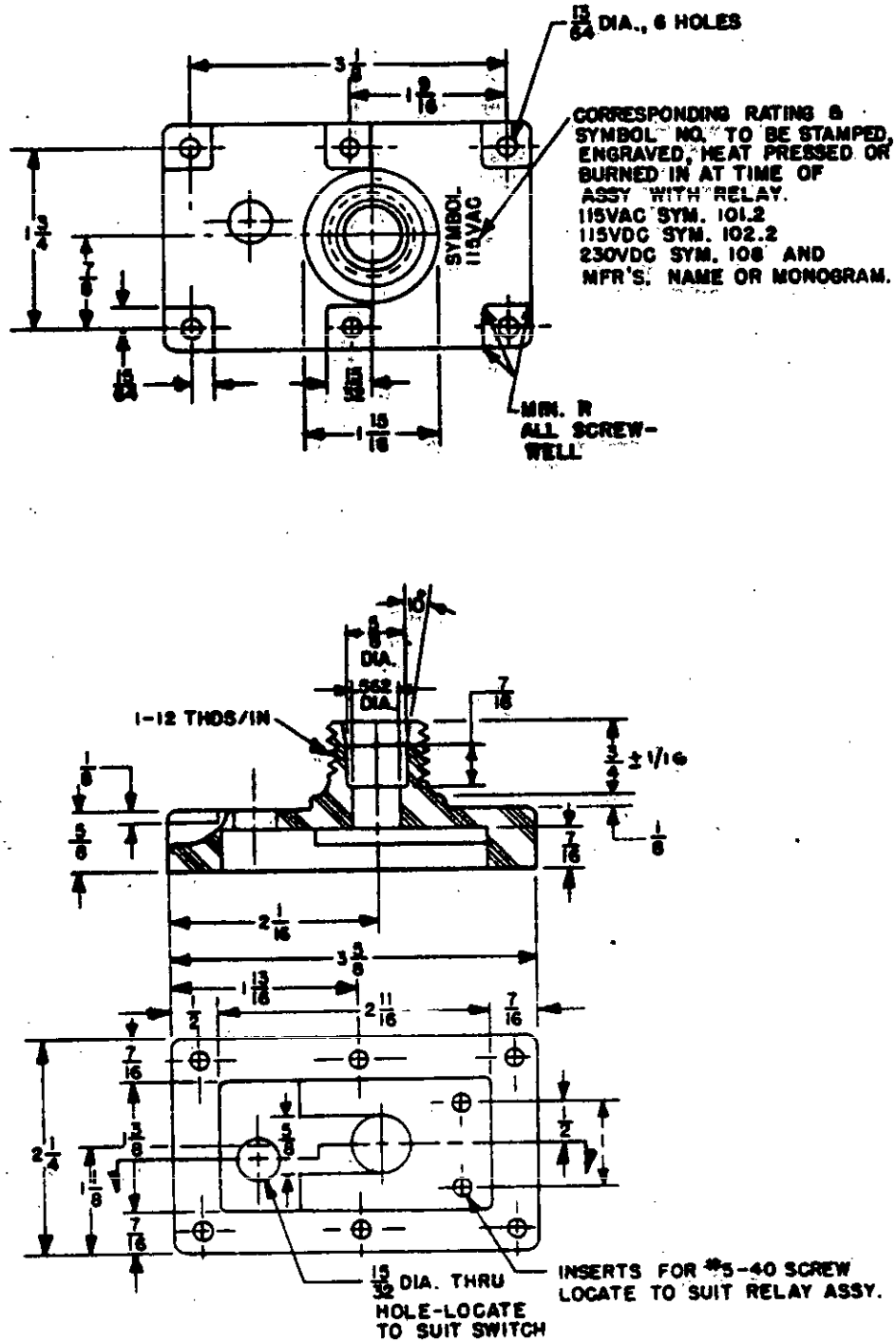
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SH 10251

Figure 10 - Handle.

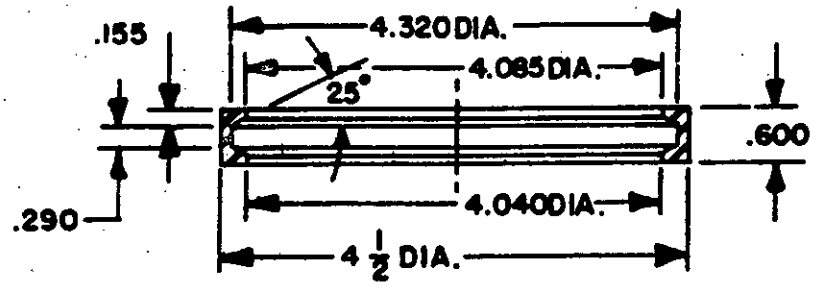
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SH 10252

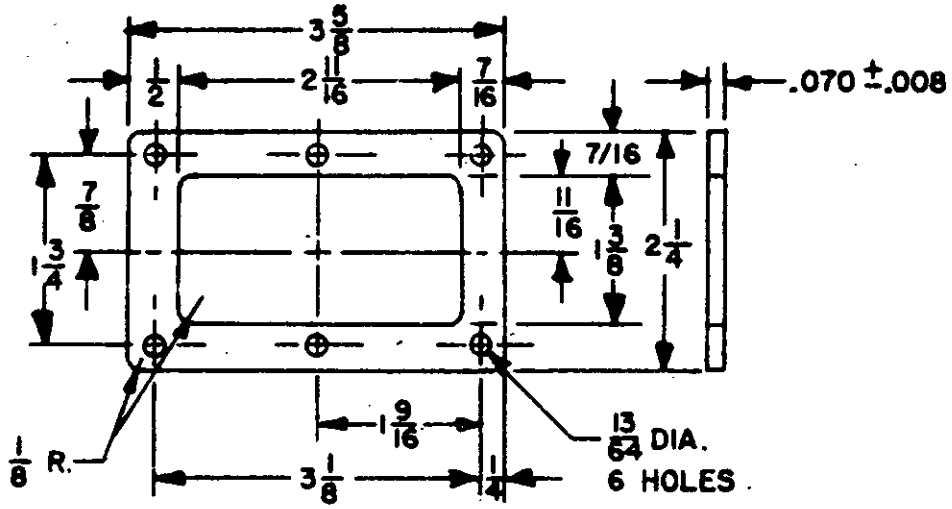
Figure 11 - Relay housing.

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SH 10253

Figure 12 - Gasket, lamp.



SH 10254

Figure 13 - Gasket handle and assembly.

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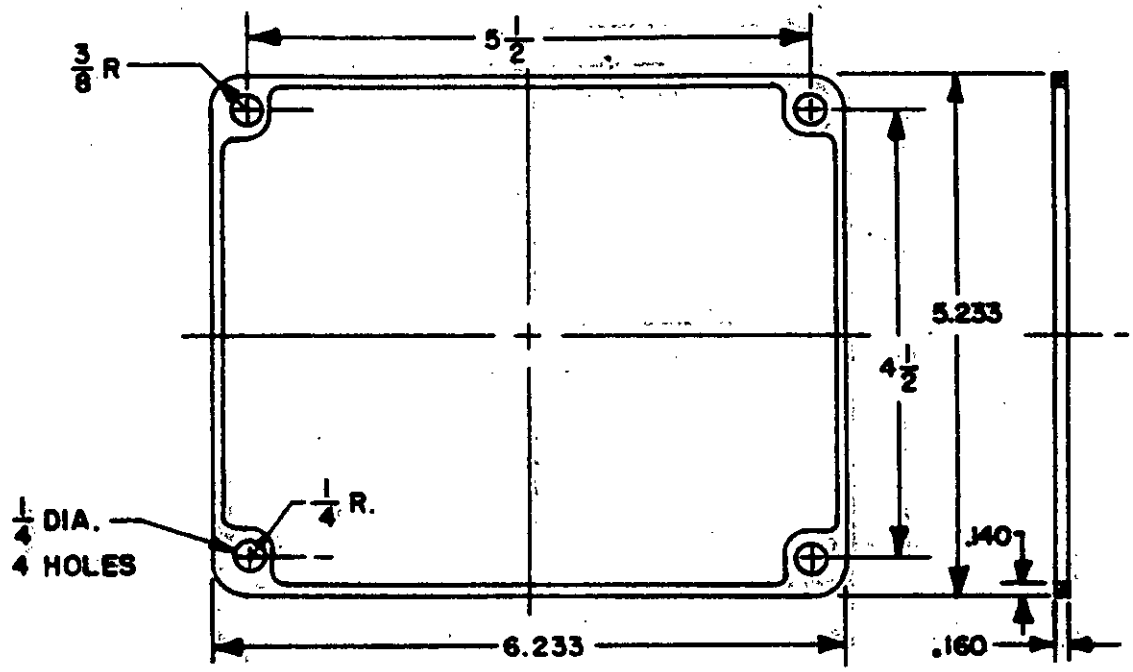
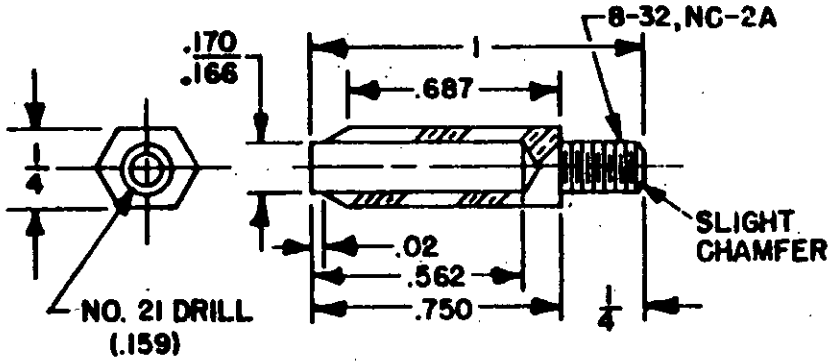
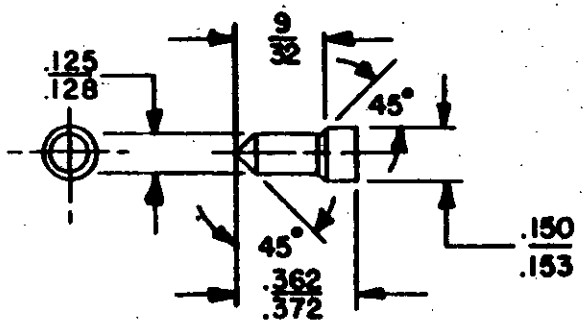


Figure 14 - Gasket, cover.

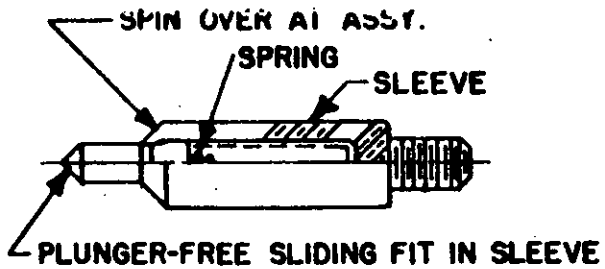
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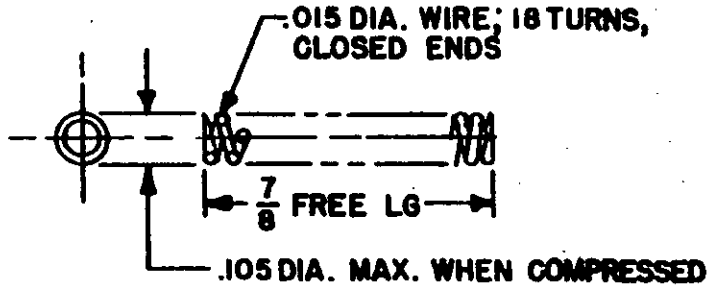
(a) SLEEVE
MATERIAL: BRASS



(b) PLUNGER
MATERIAL: CRES #304 OR BRASS



(d) ASSEMBLY

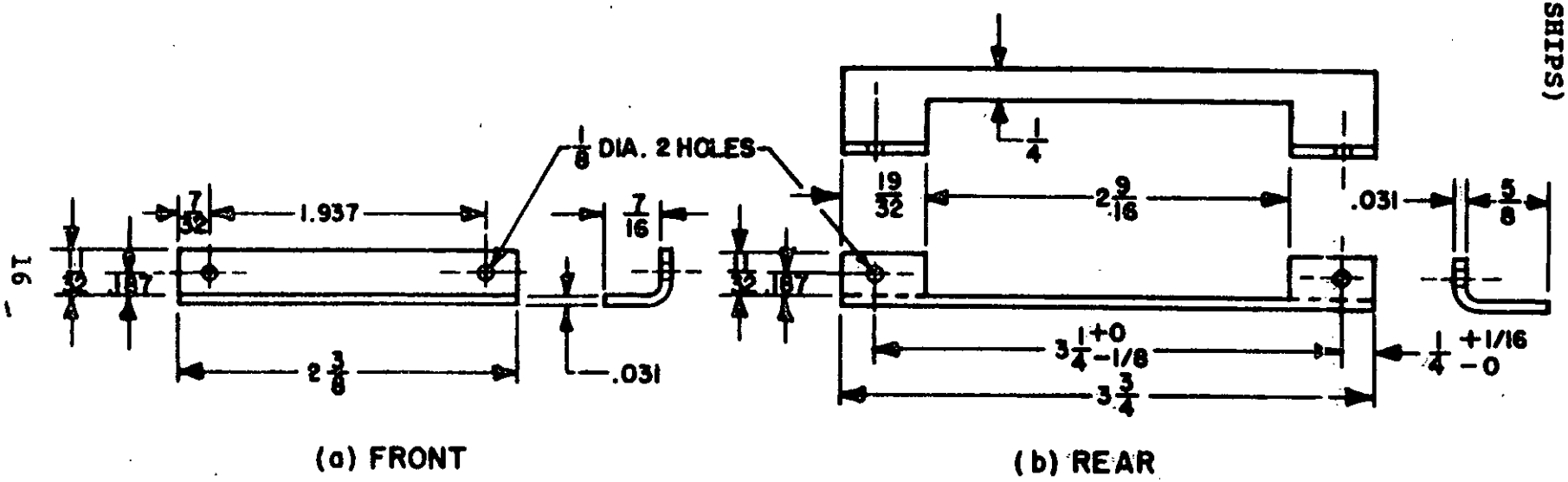


(c) SPRING
MATERIAL: PHOSPHOR BRONZE

SH 10256

Figure 15 - Lamp contact assembly.

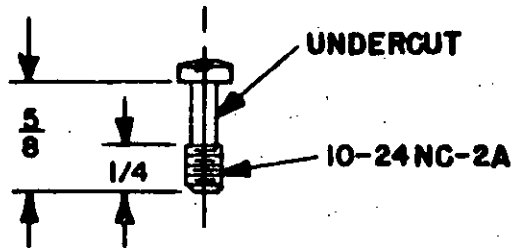
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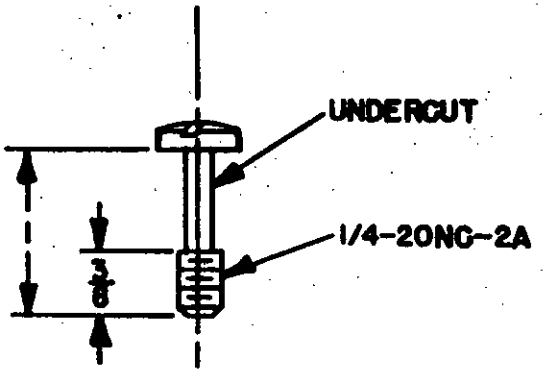
SH 10257

Figure 16 - Battery contact connectors.

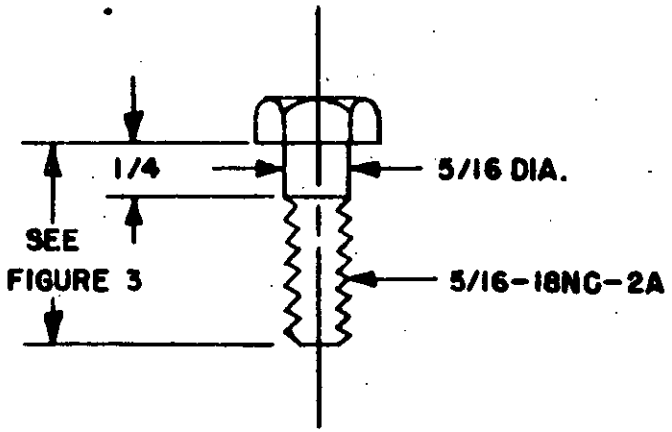
Source: https://assist.dla.mil - Downloaded: 2026-10-18T08:50Z
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(a) LAMP RETAINER SCREW
MATERIAL: BRASS



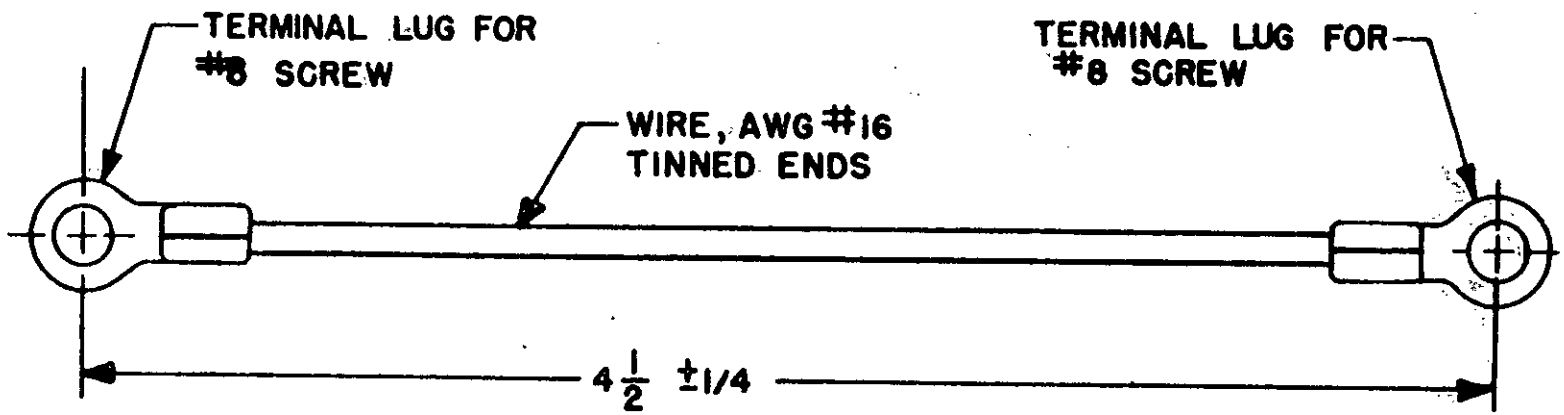
(b) COVER SCREW
MATERIAL: BRASS



(c) MOUNTING BOLT
MATERIAL: STEEL

Figure 17 - Screws and bolts.

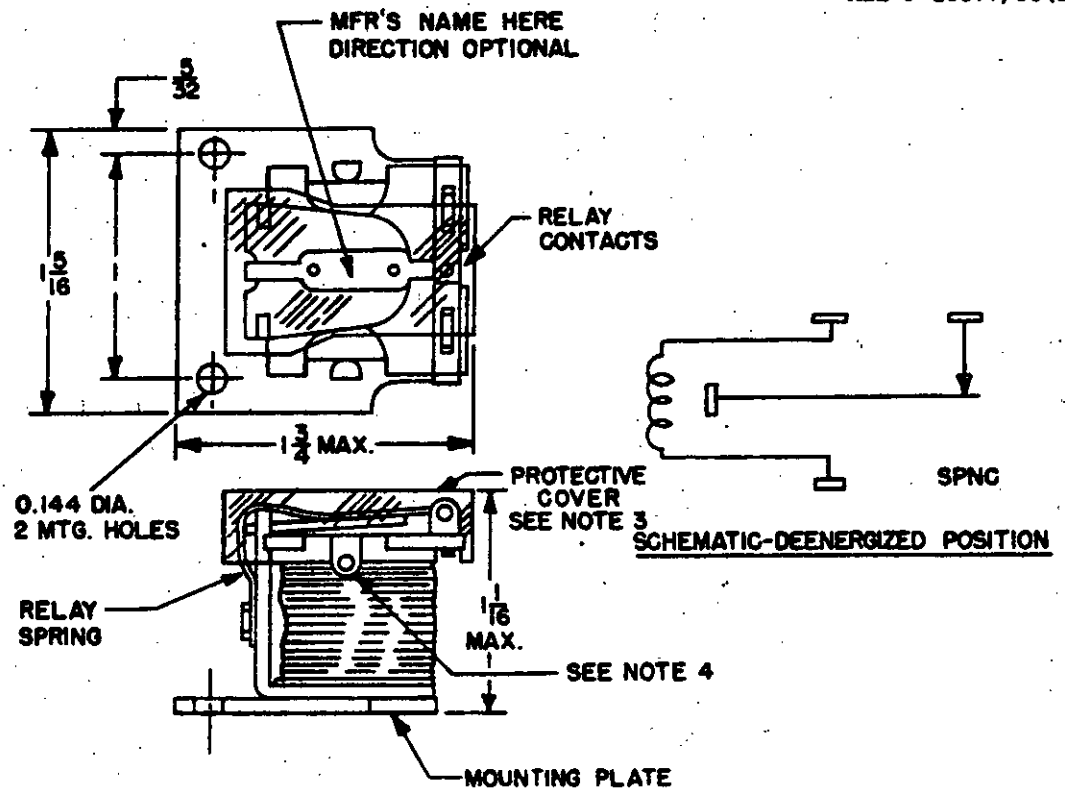
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SH 10259

Figure 18 - Wire assembly

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SH 10260

Table I - Relay contact and coil ratings.

Requirement	115 Vac	115 Vdc	230 Vdc (see note 1)
Frequency	60 Hz	---	---
Duty	Continuous	Continuous	Continuous
Max. oper. voltage	125	125	250
Coil voltage rated	115	115	230
Pull-in voltage	60-100 max.	60-100 max.	120-200 max.
Drop-out voltage	70-30 min.	70-30 min.	140-60 min.
Contact rating	2 amps, 6 volts	2 amps, 6 volts	2 amps, 6 volts
Life, cycles	100,000	100,000	100,000
Contact resistance	0.1 OHM MAX.	0.1 OHM MAX.	0.1 OHM MAX.
Insulation resistance	100 Megohms Min.	100 Megohms Min.	100 Megohms Min.
Dielectric strength	1000 V rms	1000 V rms	1000 V rms
Coil resistance	3000 OHMS	12,000	See note 1

NOTES:
1. For 230 Vdc operation, the 115-Vdc relay with 12,000 OHM, 2 watt resistor in series with coil may be used:
2. Relays shall be Frederick Controls Division part No. 1001-122D (115 VAC) and 1001-123D (115 Vdc) or equal. Substitution of equal relays shall be obtained in writing, from the Naval Ship Engineering Center.
Equivalent relays shall conform to the design shown and to the following tests:
(a) Insulation resistance - 100 MEGOHMS minimum between coil and contacts and to ground (moving contact grounded).
(b) Dielectric - 500 Vac rms between contacts, 1000 Vac rms between coil and contacts and coil to ground.

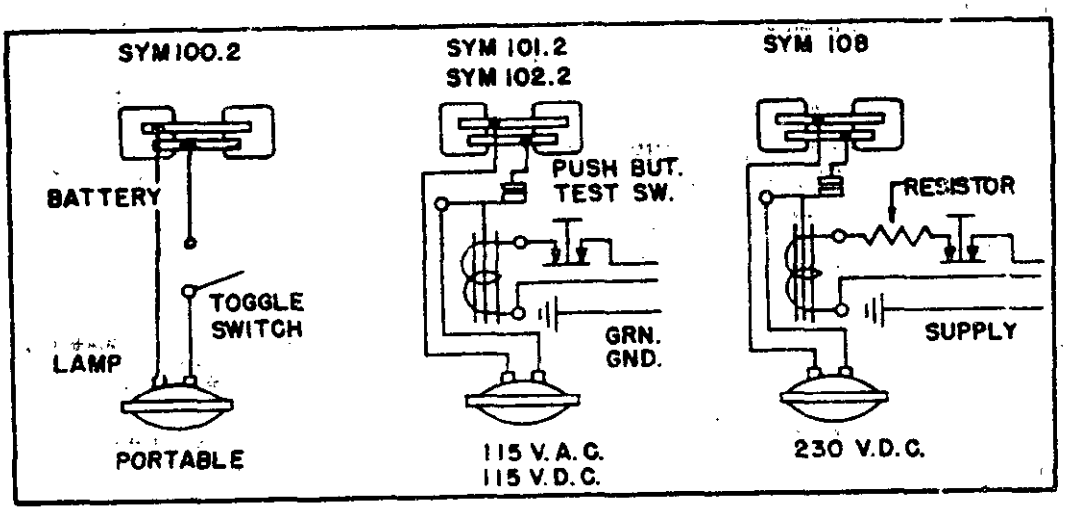
Figure 19 - Relay.

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(c) Contact resistance - 0.1 OHM maximum.
(d) Contact and coil ratings shall be as specified in table I.
(e) Vibration - relays shall be tested for type I vibration in accordance with MIL-STD-167. The relay shall be energized and de-energized for 1 hour each in each of three mutually perpendicular planes.
(f) Shock - relays shall withstand the shock test for grade A, type A, equipment in accordance with MIL-S-901, without contact chatter in excess of 100 microseconds.
(g) Salt spray - relays shall be tested in accordance with method 101, test condition B, or MIL-STD-202. After test, relay shall be washed, shaken and air blasted and then allowed to dry for 24 hours. Relays shall then be examined for corrosion and exposure of base metal.
(h) Moisture resistance - relays shall be exposed to a 100 hour humidity test as follows: Assemblies shall be placed in a chamber at a temperature of 50°C. at a relative humidity of 90 to 100 percent (dew point reached at least once each 24 hours). A 225 +25 volt source of dc shall be provided. One coil terminal shall be connected to the positive source and the negative shall be attached to the frame. After the test, coil resistance shall still be as required.
(i) The relay shall function satisfactorily in an ambient temperature varying from -30° to +160°F.

3. A protective cover shall be provided to protect the relay contacts and the relay spring from jamming against the relay wiring and relay wall cavity.
4. Relays may be furnished with 2 inch wire leads, AWG No. 20 or larger, stranded, thermoplastic, for connecting the relay coil to the supply cable and to switch off the relay assembly. Shrink-on insulation sleeves shall be placed over the 120 volt and over electrical connections in such a manner that no metallic or conducting parts shall be exposed.

Figure 19 - Relay (cont'd).



SH 10261

Figure 20 - Wiring diagram.

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REQUIREMENTS:

Type:	II
Class:	1
Symbol 100.2:	Lantern, portable, with handle and mounting bracket.
Symbol 101.2:	Lantern, relay, 115 VAC.
Symbol 102.2:	Lantern, relay, 115 VDC.
Symbol 108:	Lantern, relay, 230 VDC
Design:	
Dimensions and configurations:	See figures 1 through 19.
Material:	
Box, cover, lamp retainer and relay housing:	Material for box, cover lamp retainer and relay housing shall be molded plastic. The color of molded parts shall be yellow conforming to color No. 13655 of FED-STD-595 attained without resort to painting, enameling or other post-molding applications. Plastic material shall be type MAI-60 of MIL-M-14 or one of the following: Fiberite No. X5064, Glaskyd No. 1902, Plaskon No. 446, Raflon No. 1030, or equal.
Screws:	All screws shall be brass.
Lockwashers:	All lockwashers shall be phosphor-bronze, internal-lock.
Inserts:	All inserts shall be brass.
Lantern assembly (complete) and typical mounting:	See figures 1 and 2.
Body assembly:	See figure 3. All parts shown on figure 3 shall be furnished assembled as indicated.
Bolts:	Gaskets or bolts will self-contained gaskets such as "steel-bolts" shall be furnished so as to seal the mounting holes against the entrance of water when the body is assembled and mounted in the condition shown on figure 2.
Mounting bolts:	Mounting bolts shall be as shown on figures 3 and 17.
"O" ring:	"O" ring gaskets shall be furnished to seal the mounting holes against the entrance of water when the body is assembled and mounted in the condition shown on figure 2.
Lamp:	Lamp shall be incandescent, sealed beam, PAR 36, Industry Number 4546, FSN 6240-866-4143 conforming to MS16524-2. Lamp shall be furnished.
Lamp contact assembly:	See figure 15. The plunger shall work freely in the sleeve without rubbing or coming adrift.
Battery contact connectors:	See figure 16. Material shall be brass, half hand. Battery contact connectors shall be silver plated.
Box, cover and lamp-retainer:	See figures 7, 8, and 9 respectively.
Wire assembly:	See figure 18.
Gasket, lamp:	See figure 12. Material shall be neoprene rubber, 45-55 durometer.
Gasket, cover:	See figure 14. Material shall be neoprene rubber, 45-55 durometer.
Handle assembly:	See figure 4. All parts shown on figure 4, shall be furnished assembled as indicated.
Switch:	Switch shall be toggle, DPST, FSN 5930-088-4675 conforming to MS16656-2.
Boot:	Boot shall be type A, style 3 of MIL-B-5423/2, FSN 5975-892-9044.
Gasket:	See figure 13. Material shall be neoprene rubber 45-55 durometer.
Bracket assembly:	See figure 5. Material shall be steel; zinc or cadmium plated.

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Relay assembly:	See figure 6. All parts shown on figure 6 shall be furnished assembled as indicated.
Cable:	Cable shall be ten feet of type SJT or SJO, AWG No. 16, three conductor, (white, black, green) in accordance with J-C-580.
Cap:	Cap shall be in accordance with MIL-S-19622/10-002 of MIL-S-19622/10.
Packing assembly:	Packing assembly shall be in accordance with MIL-S-19622/17-02 of MIL-S-19622/17.
Switch:	Switch shall be pushbutton, SP-NC, double break, snap in accordance with MIL-S-8805/20, FSN 5930-952-9031, except that it shall be one circuit furnished without the nut and lock-washer.
Boot:	Boot shall be type B, style 2, size 3 of MIL-B-5423/7, FSN 5975-823-0392.
Relay housing:	See figure 11.
Gasket:	See figure 13. Material shall be neoprene rubber 45-55 durometer.
Relay:	See figure 19.
Wire assembly:	See figure 18.
Ground connection:	The metal neck of the pushbutton switch and the relay metallic frame shall be electrically connected to the green lead of the supply cable for grounding to ships structure.
Wiring:	Interconnecting and power leads shall be soldered on switch and relay terminals. The connections shall be mechanically secure and electrically continuous after soldering.
Insulating sleeve:	Shrink-on insulating sleeves shall be installed over all electrical connections carrying 120 or higher voltage. No metallic parts of such connections shall be exposed. Insulating sleeves shall not be loose or allowed to move. Insulating sleeves are not required for ground connections or electrical connections of the battery circuit.
Batteries:	Lanterns are designed to accommodate two batteries in parallel, type BA-200/U of MIL-B-18/34, FSN 6135-050-3280. Batteries are not furnished.
Enclosure:	Watertight-complete lanterns.
Weight:	
Symbol 100.2:	5 lbs. 9 oz. maximum without batteries.
Symbols 101.2, 102.2 and 108:	6 lbs. 8 oz. maximum without batteries.
Labels:	Labels shall be furnished in accordance with MIL-P-16377. The following labels and information shall be furnished:

Type I - See MIL-F-16377
Type III - Wiring diagram of figure 20.
Type IV - "USE FOLLOWING REPAIR PARTS

Lamp	FSN 6240-866-4143
Boot for toggle switch	FSN 5975-892-9044
Boot for push-button switch	FSN 5975-823-0392
Toggle switch	FSN 5930-088-4675
Pushbutton switch	FSN 5930-952-9031"

Quality assurance:	Quality assurance shall be as specified in MIL-P-16377 and as follows:
First article inspection:	
Body assembly:	See table II
Handle assembly:	See table II
Bracket assembly:	Examination only
Relay assembly:	See table III
Quality conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-P-16377
Examination for preparation of delivery:	See MIL-P-16377
Quality conformance tests:	See MIL-P-16377

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Table II - First article and comparison inspection for body and handle assembly.

Inspection	Requirement paragraph	Inspection paragraph	Remarks 1/
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight

1/ For test purposes, body or handle assembly shall be assembled and mounted as shown on figure 2 for symbol 100.2. Batteries shall be installed for the operation, shock and vibration tests.

Table III - First article and comparison inspection for relay assemblies.

Inspection	Requirement paragraph	Inspection paragraph	Remarks 1/
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Grounding circuit	3.6.13.3	4.8.11	

1/ Relay assmeblies shall be assembled and mounted as shown on figure 2. Batteries shall be installed for the operation, shock and vibration tests.

General information:

(a) Assignment of Military part numbers and general information is shown in table IV.

Table IV - Part number and general information.

Military part number M16377/53	Item name	Federal stock number (FSN)	Supersedes part number	Remarks
-100.2	Complete lantern, symbol 100.2	---	MS17231-100.2	1/, 5/
-101.2	Complete lantern, symbol 101.2	---	MS17231-101.2	2/, 5/
-102.2	Complete lantern, symbol 102.2	---	MS17231-102.2	3/, 5/
-108	Complete lantern, symbol 108	---	MS17231-108	4/, 5/
001	Body assembly	6230-783-6519	MS17231-1	
002	Handle assembly	6230-776-5920	MS17231-2	
003	Bracket assembly	6230-968-7831	MS17231-25	
004	Relay assembly, 115 VAC	6230-783-6251	MS17231-3	
005	Relay assembly, 115 VDC	6230-783-6252	MS17231-4	
006	Relay assembly, 230 VDC	6230-783-6253	MS17231-5	
007	Cover gasket, figure 14	---	MS17231-12	

See footnotes on next page.

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1/ Symbol 100.2 consists of part numbers: M16377/53-001, M16377/53-002 and M16377/53-003.
2/ Symbol 101.2 consists of part numbers: M16377/53-001 and M16377/53-004.
3/ Symbol 102.2 consists of part numbers: M16377/49-001 and M16377/49-005.
4/ Symbol 108 consists of part numbers: M16377/53-001 and M16377/53-006.
5/ Lanterns are not furnished as complete units. Complete lanterns shall be assembled using the component parts indicated.

(b) Lanterns are intended to be mounted onboard ship as shown on figure 2.
(c) Lanterns over surgical tables onboard ship are intended to be mounted using the adjustable bracket shown on MIL-P-16377/55.
(d) Color filters, MIL-P-16377/46, shall be used to convert white (clear) light lanterns into colored light lanterns.
(e) Superseding data: These lanterns were formerly covered by MS17231 and Drawing 815-74434 and 9-[illegible]-5293-L.

Preparing activity:
Navy - SH
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